

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $5x^5 - 6x^4 + 8x^3$?

Answer

- A. $x^4(5x - 6)$
- B. $x^3(5x^2 - 6x + 8)$
- C. $8x^3(5x^2 - 6x + 1)$
- D. $6x^5(-6x^4 + 8x^3 + 1)$

Correct Answer: B

Rationale

Choice B is correct. Since x^3 is a common factor of each term in the given expression, the expression can be rewritten as $x^3(5x^2 - 6x + 8)$.

Choice A is incorrect. This expression is equivalent to $5x^5 - 6x^4$.

Choice C is incorrect. This expression is equivalent to $40x^5 - 48x^4 + 8x^3$.

Choice D is incorrect. This expression is equivalent to $-36x^9 + 48x^8 + 6x^5$.